Huawei Access Node Optical Transport Failure Resolution SOP (JAY-ACC-77)

Node: JAY-ACC-77 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_2453

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) in the Huawei Access Node (Role: Access, Node ID: JAY-ACC-77, Site Location: Jayanagar Metro Site) is a critical failure domain that warrants immediate attention. This failure is primarily attributed to the degradation of SFP/XFP transceivers or fiber cuts, leading to a loss of receive or transmit power. The Huawei Access Node, being an optical transport equipment, relies heavily on the SFP/XFP transceivers to establish and maintain optical connections with neighboring nodes. The Rx/Tx power loss is typically detected by the node's optical monitoring system, which triggers an alarm (ALARM_2453 with Severity: Warning) to indicate a potential issue. To troubleshoot this failure, it is essential to understand the underlying protocol and hardware mechanics. The Huawei Access Node uses the SONET/SDH protocol to establish and manage optical connections. The SFP/XFP transceivers are responsible for converting electrical signals to optical signals and vice versa. When an SFP/XFP degrades or a fiber cut occurs, the receive or transmit power is affected, leading to a loss of signal quality or complete loss of connection. The diagnostic and resolution steps outlined in this SOP are designed to identify the root cause of the failure and apply corrective measures to restore normal operation.

Pre-requisite Checks

- Verify the site location (Jayanagar Metro Site) and node ID (JAY-ACC-77) to ensure the correct hardware is being accessed.

- Check the redundant link status to ensure a backup path is available in case of a permanent failure.

- Dispatch remote hands to the site location for assistance (if required).

- Review the alarm history and system logs to identify any recent issues or maintenance activities that may have contributed to the failure.

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation	High
Fiber cuts or damage	Medium
Rx/Tx power loss due to misconfiguration or hardware issues	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-channel, Expected Output: Optical Channel
Information Node ID: JAY-ACC-77 Optical Channel: OC-48 Rx Power: -20
dBm Tx Power: -15 dBm, Description: This command displays the optical
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channel information, including the node ID, optical channel details, receive power, and transmit power. The receive power and transmit power values can be used to identify any issues with the SFP/XFP transceivers or fiber connections.

Command: display sfp, Expected Output: SFP Information SFP Type: SFP-1GE-LX SFP Status: Degraded SFP Temperature: 45°C, Description: This command displays the SFP information, including the SFP type, status, and temperature. A degraded SFP status indicates a potential issue with the SFP transceiver.
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Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP transceiver, Command: replace sfp 1/1-1, Description: Replace the degraded SFP/XFP transceiver with a new one to restore normal operation. Ensure the new SFP/XFP transceiver is properly seated and secured.

Step 2: Step: Verify the fiber connections and repair any damage, Command: display optical-channel, Description: Verify the fiber connections and repair any damage to restore normal operation. Use the display optical-channel command to check the receive and transmit power values.

Post-Check Verification

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Command: display optical-channel, Expected Output: Optical Channel Information Node ID: JAY-ACC-77 Optical Channel: OC-48 Rx Power: -20 dBm Tx Power: -15 dBm, Description: Verify that the receive power and transmit power values are within normal limits to ensure the SFP/XFP transceivers and fiber connections are functioning correctly.

Command: display sfp, Expected Output: SFP Information SFP Type: SFP-1GE-LX SFP Status: Normal, Description: Verify that the SFP status is normal to ensure the SFP transceiver is functioning correctly.
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Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: restore sfp 1/1-1, Description: Restore the original SFP/XFP transceiver to its previous state if the replacement did not resolve the issue.

- Step: Verify the fiber connections and repair any damage, Command: display optical-channel, Description: Verify the fiber connections and repair any damage to restore normal operation. Use the display optical-channel command to check the receive and transmit power values.